

| Year | JDK version        | Features                                                                                                                                                                                                |
|------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1996 | JDK 1.0            | Initial release of Java. Revolutionary for its time with applets, AWT, basic libraries and JVM                                                                                                          |
| 1997 | JDK 1.1            | Inner classes, anonymous classes, New event delegation model, JavaBeans, JDBC (Java DataBase connectivity), Reflection API, Deprecated Thread methods (stop(), resume(), suspend())                     |
| 1998 | J2 SE 1.2 (Java 2) | Swing GUI toolkit, collections framework, Just-In-Time compiler improvements, Strictfp keyword, security enhancement                                                                                    |
| 2000 | J2 SE 1.3          | Java Naming & Directory Interface (JNDI) in core, Hotspot JVM, JavaSound API, RMI over IIOP.                                                                                                            |
| 2002 | J2 SE 1.4          | assert keyword, chained Exceptions, NIO (New Input/output Package), Logging API, regular expressions (Pattern & Matcher), XML processing.                                                               |
| 2004 | J2 SE 5 (1.5)      | Major Revolutionary upgrade: Generics, Annotations, Autoboxing / unboxing, Enums, Enhanced for-each loop, var args, Static import, formatted I/O (Printf), Concurrency utilities (java.util.concurrent) |
| 2006 | Java SE 6 (1.6)    | Integrated web services (JAX-WS), compiler API (JSR 199), Improvements to JDBC 4.0, GUI improvements (Swing Worker, table sorting), scripting support (JSR 223).                                        |

2011

Java SE 7 (1.7)

Project coin Features:

Strings in switch, Diamond operator (<>), Try-with-resources (AWT), Multicatch exception handling, binary literals, underscores in numeric literals, Fork / Join Framework for parallelism, NIO.2 (Enhanced file system and I/O APIs)

2014

Java SE 8 (1.8)

Massive upgrade: Lambda expressions, Stream API, functional interfaces, default and static methods in interfaces, New Date & Time API, parallel array sorting, type annotations.

2017

Java SE 9

Factory methods for List, Set, Map

Modules system (Project Jigsaw), jshell (REPL tool), private interface methods, New @index tag in javadoc with search, deprecation of Applet API, jlink tool to create custom runtime images.

2018 (March)

Java SE 10

Local variable type inference (var), New version string scheme, performance improvements (Garbage Collector improvements)

2018 (Sept)

Java SE 11 (LTS)

Long Term Support

var in lambda parameters, HTTP Client API (java.net.http), removal of applets and Java web start, Removal of JavaFX from JDK, Direct execution of single file programs.

2019 (March)

Java SE 12

Preview of switch expressions (value-returning switches)



|              |                     |                                                                                                                                                                            |
|--------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2019 (Sept)  | Java SE 13          | Preview of text blocks (multi-line strings)                                                                                                                                |
| 2020 (March) | Java SE 14          | Final switch expressions, Pattern matching for instanceof (Preview)                                                                                                        |
| 2020 (Sept)  | Java SE 15          | Final text blocks, sealed classes and interfaces (Preview)                                                                                                                 |
| 2021 (March) | Java SE 16          | Records (data carrier classes), Pattern matching for instanceof (finalized), jpackage tool for packaging applications                                                      |
| 2021 (Sept)  | Java SE 17<br>(LTS) | Sealed classes and interfaces (final), Context-sensitive keywords: sealed, permits, non-sealed; Deprecation of Applet API for removal.                                     |
| 2023 (Sept)  | Java SE 21<br>(LTS) | Pattern Matching in switch, Record patterns, sequenced collections, Virtual threads (lightweight concurrency), Preview features: String Templates, Structured concurrency. |

### Major Milestones :-

- JDK 5 : Generics, Annotations, enums, concurrency utilities
- JDK 8 : Lambdas and Streams → Functional programming
- JDK 9 : Module System (Project Jigsaw).
- JDK 10 : var for local variables
- JDK 17 : Sealed classes, modernized design.
- JDK 21 : virtual threads & advanced pattern Matching.